Impact of Industrial Waste on A Lake Ecosystem

Conceptual and Mathematical MODELLING of phosphorous waste

Alexis Kuerbig, Rachel Leung, Niko Vahlas, Cindy Wong, Ruoxuan Li

Introduction

- PHOSPHORUS: Key nutrient for crops, growth enhancement

- Runoff into waterways introduces phosphorus into aqueous ecosystems
  - Phosphorus causing **enhanced algae growth = reduced oxygen + death of biotic systems**
  - Hazardous to aquatic life, risk to human health → drinking water & food sources

- Degree of risk changes over time with respect to total phosphorus content in (aqueous) reservoir
  - **Consumer demand** for industrial product: Phosphorus waste enters aqueous ecosystems → harms aquatic life
  - **Construction projects** increase risk of phosphorus runoff into local waterways and introduce increased risk to ambient aquatic life

QUESTIONS:

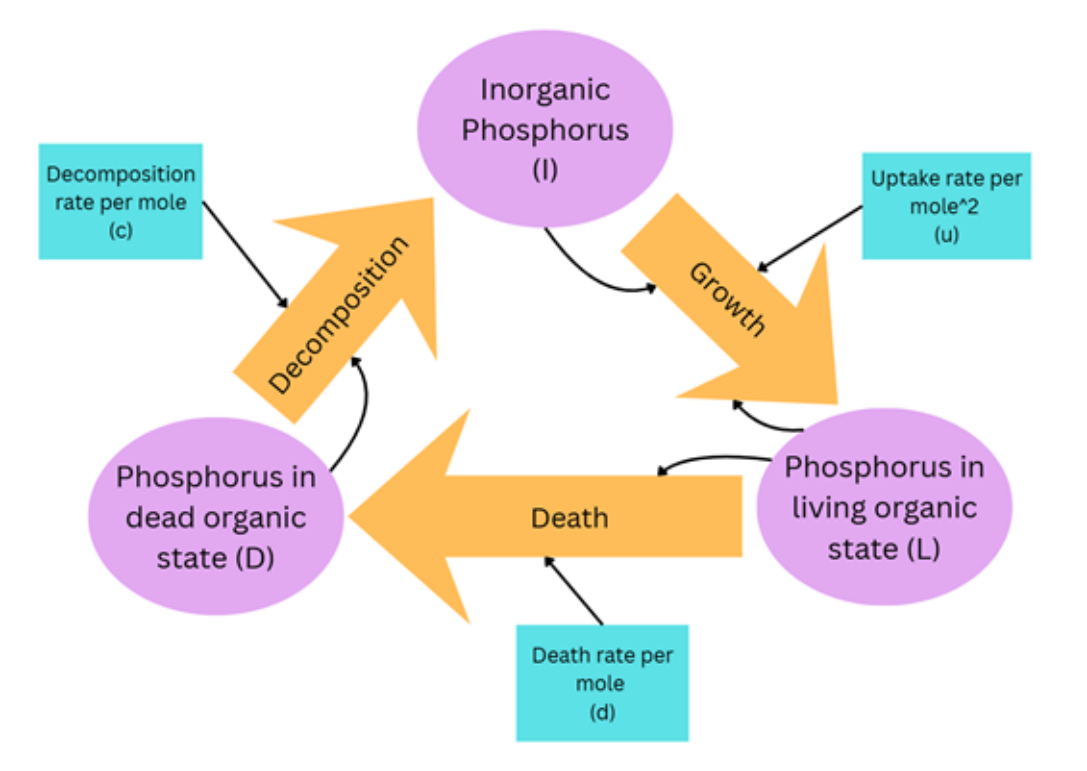
1. How can the flow of phosphorus create implications on aquatic phosphorus cycles?

2. How can modelled changes to phosphorus cycles, and associated changes to ecosystem and aquatic life, inform future construction projects and planning?

- Assume phosphorus is the only factor affecting algae growth– only contribution to changes in aqueous phosphorus cycle

https://agriculture.canada.ca/en/agricultural-production/water/water-contamination-phosphorus

Model Description



Reservoirs:

1. Phosphorus in living organic state (L)
2. Phosphorus in dead organic state (D)
3. Inorganic phosphorus (I)

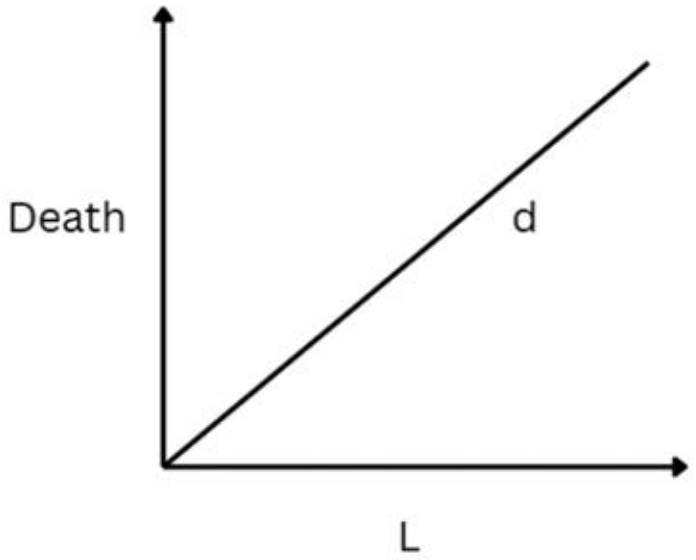
Converters:

1. Death rate per mole (d)
2. Decomposition rate per mole (c)
3. Uptake rate per mole (u)

Model Derivation

(Assumes linear relationship for all graphs)

For death rate per mole (d):



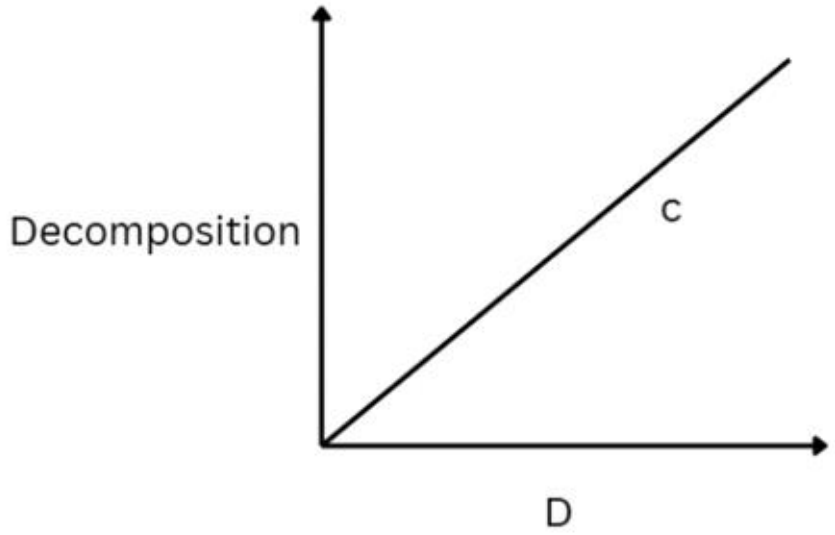
$$Death = d * L$$

$$d = \frac{Death}{L}$$

- Death of algae depends on death rate (d) and the amount phosphorus in the living organic state (L)

Model Derivation

For decomposition rate per mole (c):



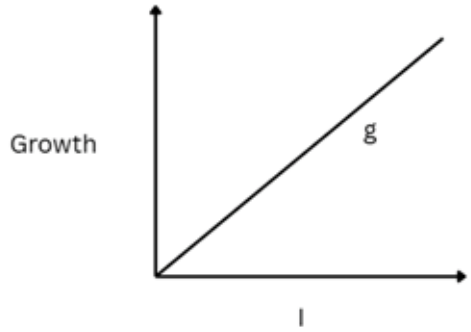
$$Decomposition = c * D$$

$$c = \frac{Decomposition}{D}$$

- Decomposition depends on decomposition rate (c) and phosphorus in dead organic state (D)

Model Derivation

For uptake rate per mole^2 (u):
Introduce new variable, growth rate (g)

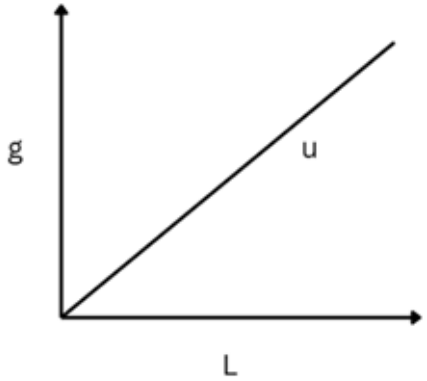


$$Growth = g * I$$

$$g = \frac{Growth}{I}$$

- Growth depends on growth rate (g) and the amount of inorganic phosphorus (I)
- Growth rate depends on the uptake rate and phosphorus in living organic state (L)

From that we get:



$$g = u * L$$

$$u = \frac{g}{L}$$

Conceptual Model becomes a mathematical model

Parameters, values

Inorganic Phosphorus (I) Reservoir:

$$\frac{dI(t)}{dt} = c * D(t) - u * L(t) * I(t)$$

With the forward difference method:

$$I(t + \Delta t) = I(t) + [c * D(t) - u * L(t) * I(t)]\Delta t$$

Phosphorus in living organic state (L) reservoir:

$$\frac{dL(t)}{dt} = u * L(t) * I(t) - d * L(t)$$

With the forward difference method:

$$L(t + \Delta t) = L(t) + [u * L(t) * I(t) - d * L(t)]\Delta t$$

Phosphorus in dead organic state (D) reservoir:

$$\frac{dD(t)}{dt} = d * L(t) - c * D(t)$$

With the forward difference method:

$$D(t + \Delta t) = D(t) + [d * L(t) - c * D(t)]\Delta t$$

- The rate of change in the reservoirs are dependant on the inflows and outflows
- Inorganic phosphorus (I)
  - inflow: decomposing dead organic phosphorus
  - outflow: growing inorganic phosphorus
- phosphorus in living organic state (L)
  - inflow: growing inorganic phosphorus
  - outflow: dying organic phosphorus in the living state
- Phosphorus in dead organic state (D)
  - inflow: dying organic phosphorus in the living state
  - outflow: decomposing dead organic phosphorus

Conceptual Model becomes a mathematical model

Parameters, values

Values for the parameters:

$$Phosphorus\ in\ the\ living\ organic\ state\ L(t) = 0.2\ moles$$
$$Inorganic\ phosphorus\ I(t) = 1.0\ moles$$
$$Phosphorus\ in\ dead\ organic\ state\ D(t) = 0.1\ moles$$

$$d = \frac{1}{4} = \frac{0.25}{(year)(mol)}$$
$$c = \frac{1}{20} = \frac{0.05}{(year)(mol)}$$
$$u = \frac{1}{0.403} = \frac{2.48}{(year)(mol^2)}$$
$$\Delta t = 0.1\ years$$

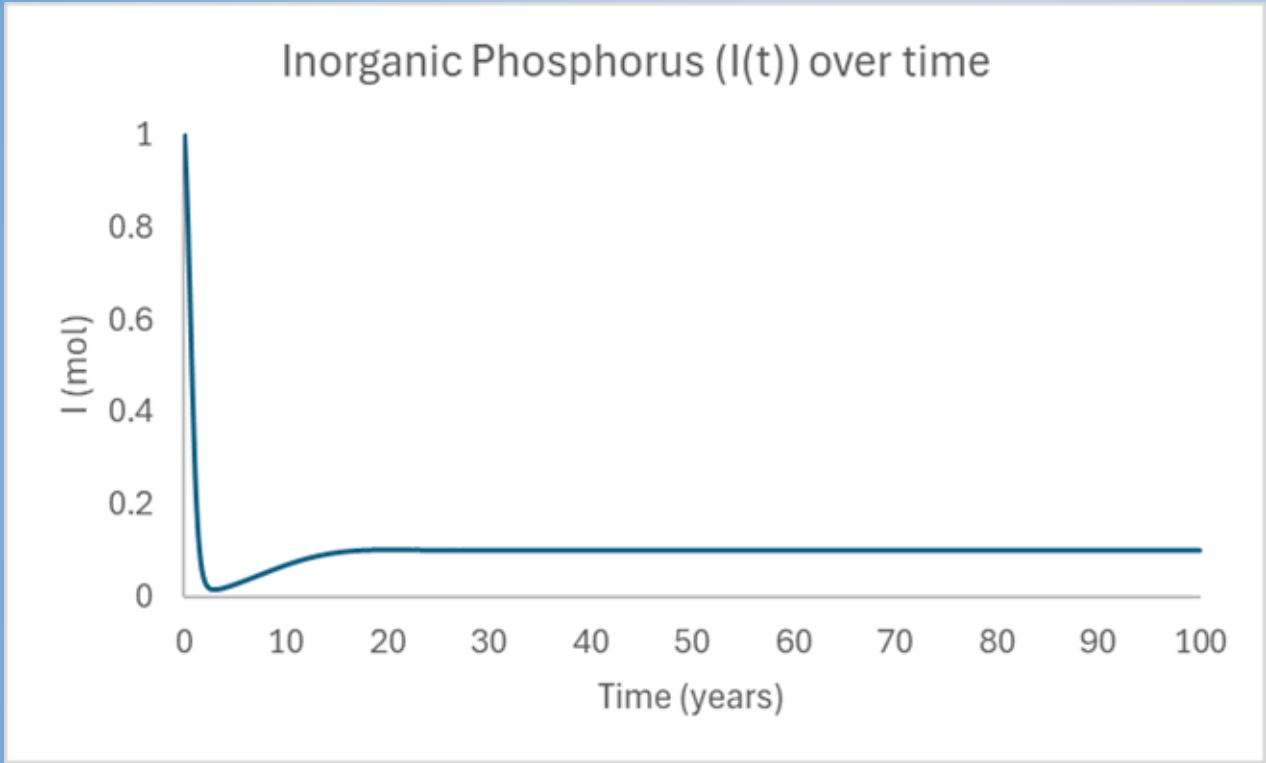
- Values obtained from the field for the lake system

Model Results

Questions:

1. For what conditions (relations among variables) each of the reservoirs can be in steady state?
2. How do the three reservoirs change over 100 years, with time step 0.1 year?
3. How sensitive are the results to changes in the parameter values of d, c, and u. And how sensitive are the results to changes in initial conditions of the reservoirs?
4. Once the reservoirs hit steady state, what happens when an influx of inorganic phosphorus is experienced over time and what happens when there is a one-step influx of inorganic phosphorus?

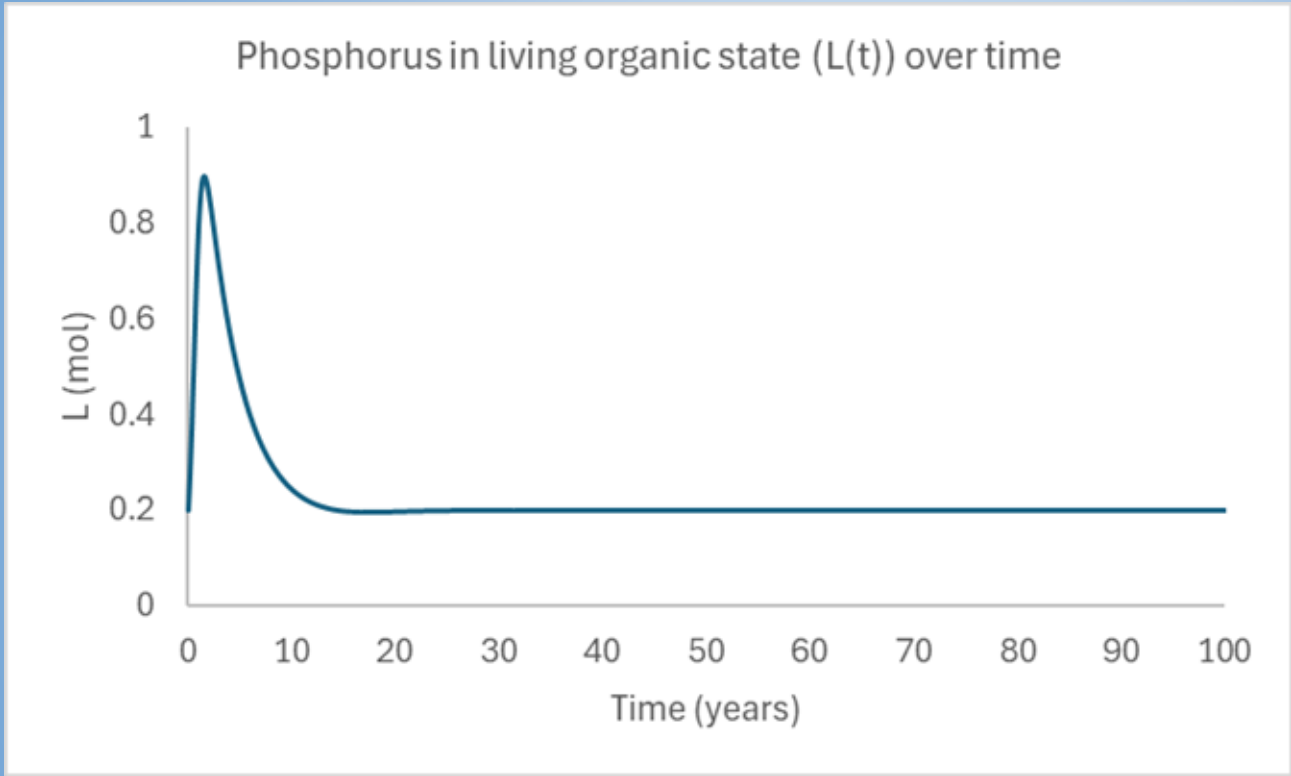
Model Results + graphical analysis



In the system, there is initially 1 mol of inorganic phosphorus, and after 32.9 years, the amount settles to 0.1008 mol and stops changing

Figure 1. Inorganic phosphorus in the lake ecosystem over 100 years.

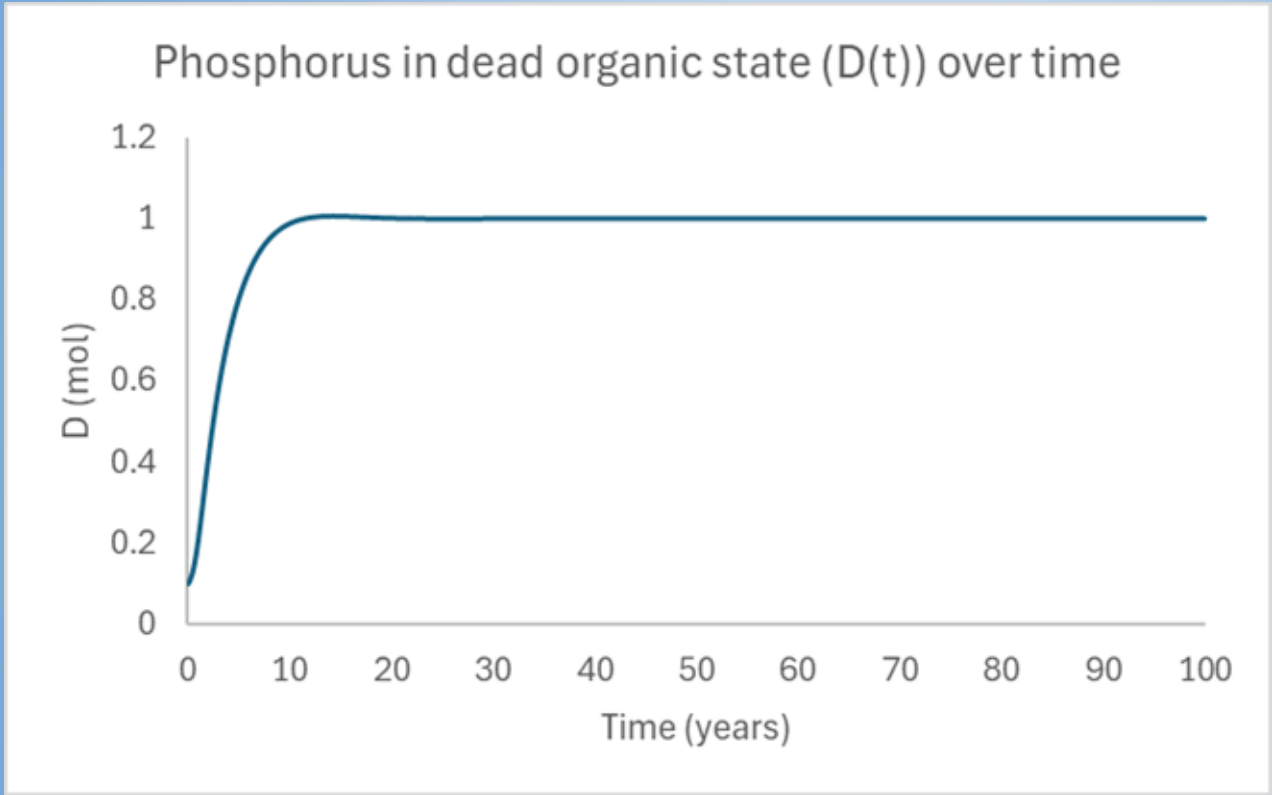
Model Results + graphical analysis



phosphorus in living organic state: We start with 0.2 mol, and after 32.5 years, the amount settles at 0.1999 mol.

Figure 2. Phosphorus in the living organic state in the lake ecosystem over 100 years.

Model Results + graphical analysis



phosphorus in the dead organic state: We start with 0.1mol and eventually after 30.8 years it reaches steady state, where there is 0.9993 mol.

Figure 3. Phosphorus in the dead organic state in the lake ecosystem over 100 years.

How sensitive are the results to changes in parameters (converters)?

- **Changes in death rate (d)**
  - Significantly changes final amounts of L(t) and I(t) but not D(t)
  - E.g. Increasing death rate by 2 doubles final I(t) and halves L(t)
  - Change to shape of graphs:

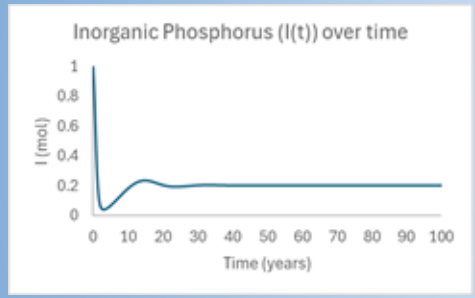


Figure 4. Inorganic phosphorus in the lake ecosystem when death rate is increased.

Figure 5. Phosphorus in living organic state in the lake ecosystem when death rate is increased.

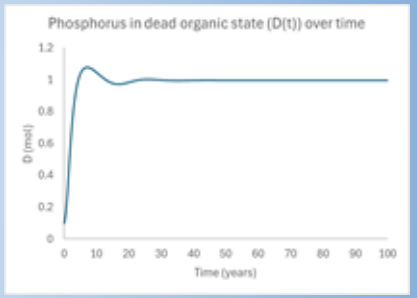


Figure 6. Phosphorus in dead organic state in the lake ecosystem when death rate is increased.

- **Changes in decomposition rate (c):**
  - Significantly changes final L(t), slightly changes D(t), and does not change I(t)
- **Changes in uptake rate (u)**
  - Significantly changes final I(t) but minimal changes in L(t) and D(t)

How sensitive are the results to changes in parameters (reservoirs)?

- Results are sensitive to changes in I(t)
  - Increasing I(t) by 2 -> increases L(t) from 0.1999 mol to 0.3665 mol and D(t) from 0.9993 mol to 1.8327 mol (around 2 times increase)
  - Decreasing I(t) by 2 -> decreases L(t) and D(t) by around 2 times

- Results are not as sensitive to changes in L(t) and D(t)
  - Increasing either -> small increases in L(t) and D(t)
  - Decreasing either -> small decreases in L(t) or D(t)

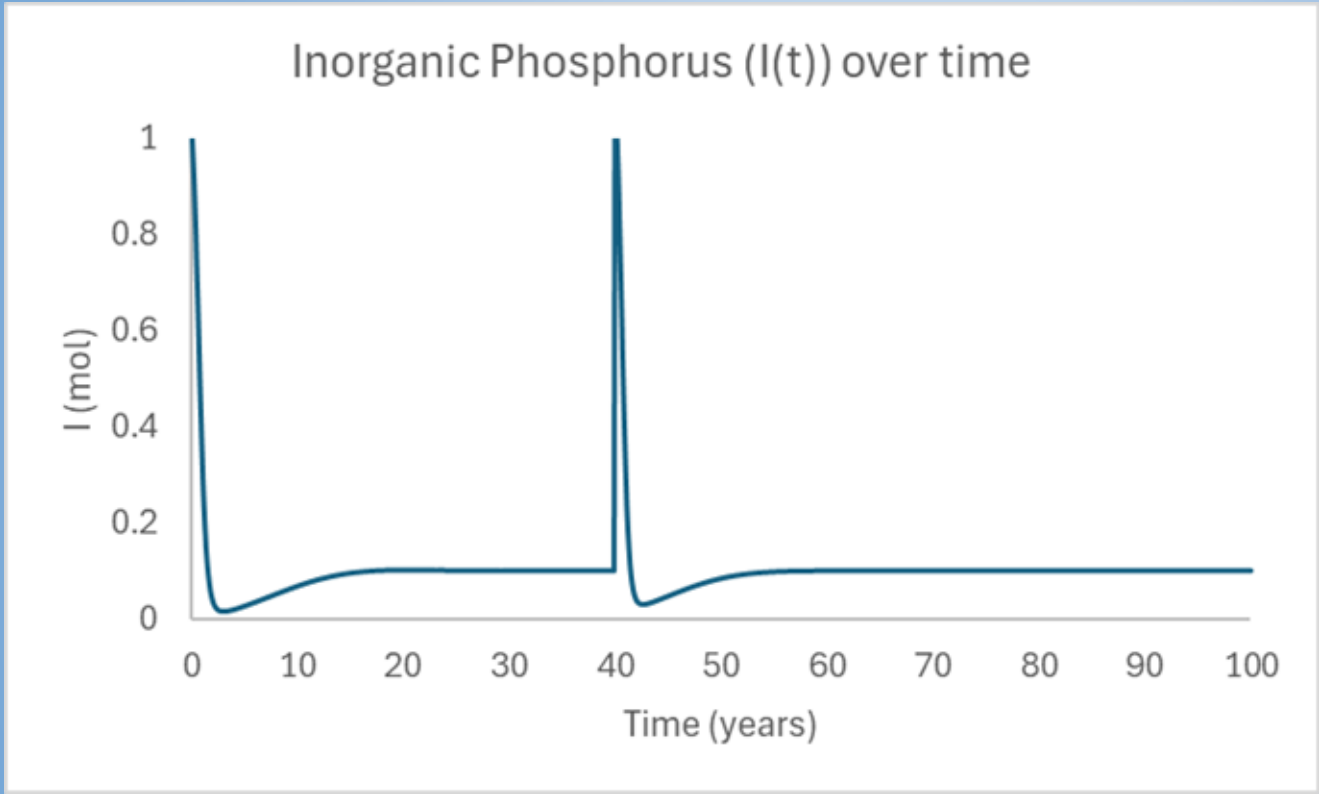
- Changing I(t), L(t) and D(t) do not change the final steady state amount of I(t)

Scenario 1

1 mol of inorganic phosphorus is inputted into the ecosystem in 1 year after the systems had reached steady state.

Year of reaching steady state: year 40.

Scenario 1

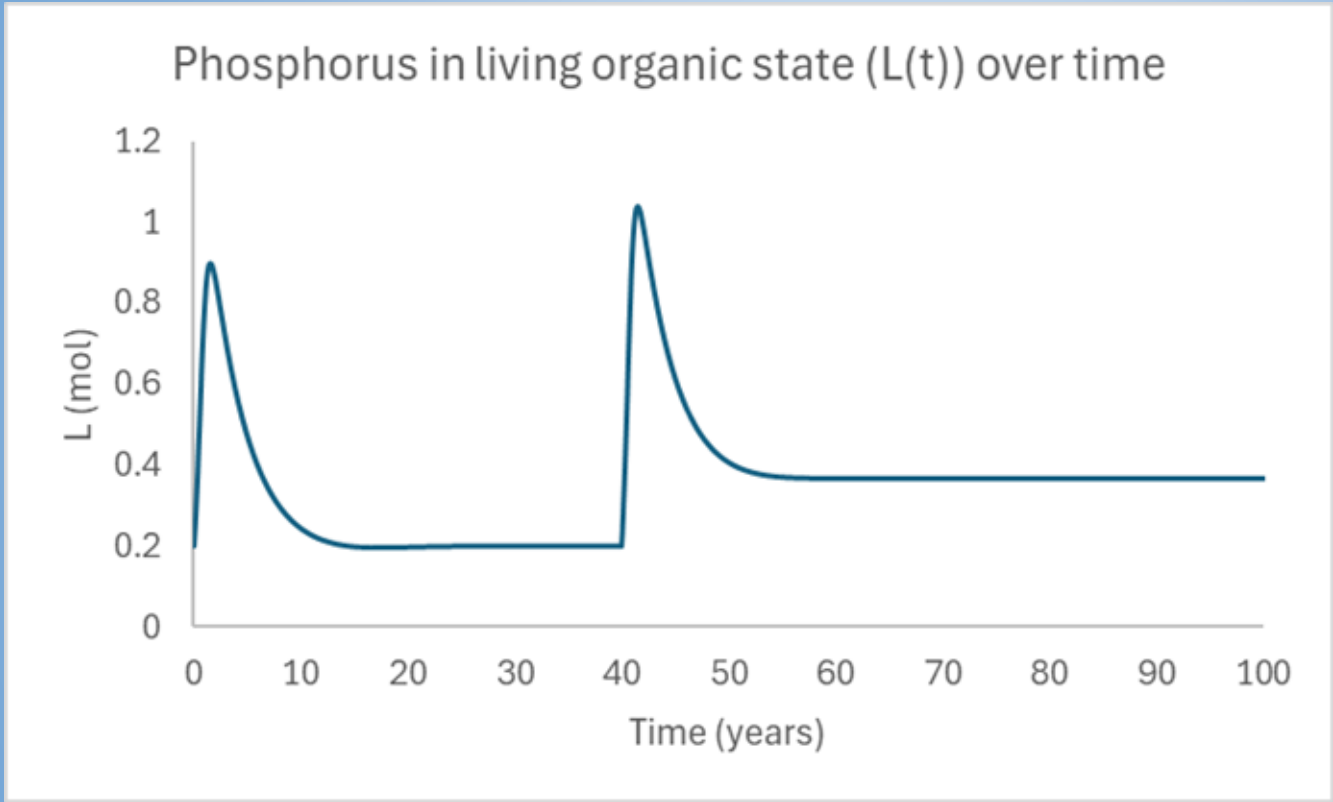


The inorganic phosphorus system reached a new steady state at 60.6 years, which is 21 years after the inorganic phosphorus was inputted.

The amount of inorganic phosphorus in the new steady state is the same as the original steady state (remains 0.1008 mol).

Figure 7. Inorganic phosphorus in the lake ecosystem over 100 years, with an input of 1 mol inorganic phosphorus at year 40.

Scenario 1

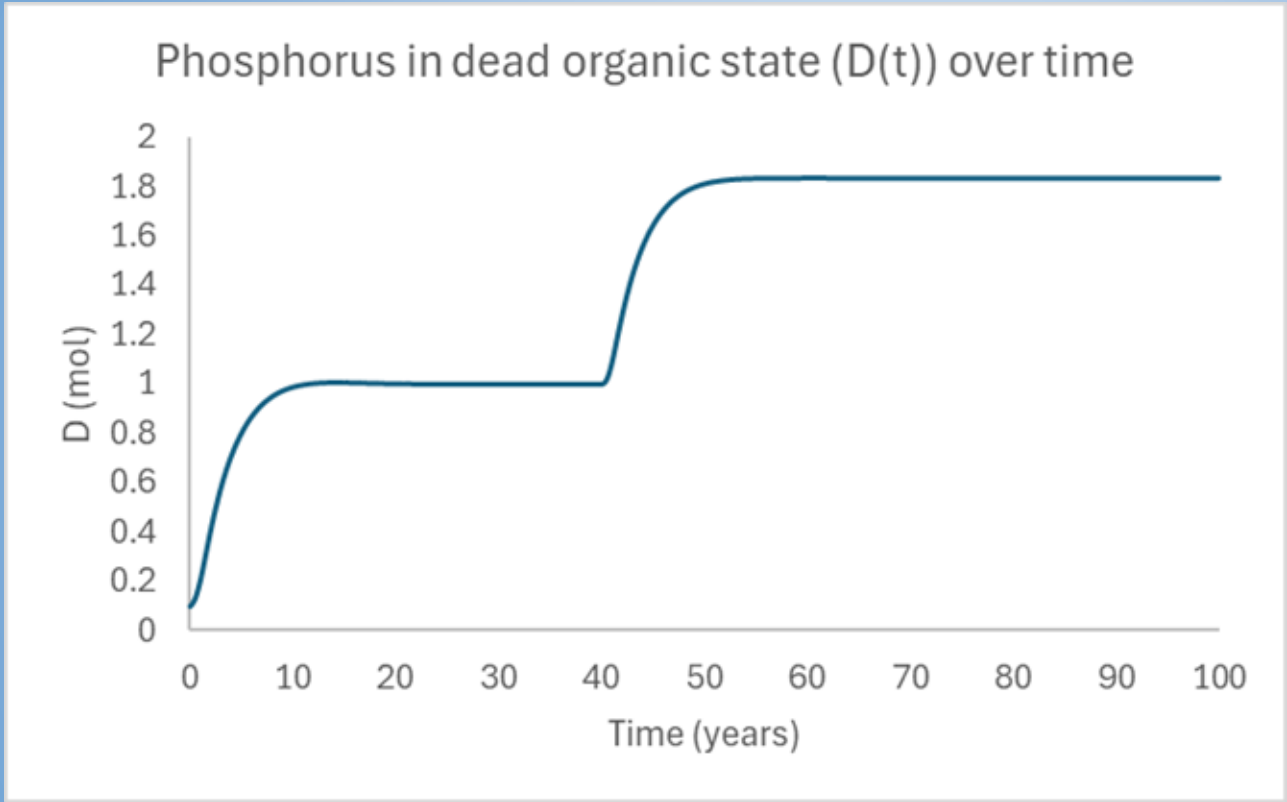


Phosphorus in the living organic state reached a new steady state at 60.3 years, which is around 20 years after the inorganic phosphorus input.

This new steady state amount is 0.3665 mol, which is greater than the original steady state value for living organic phosphorus.

Figure 8. Phosphorus in the living organic state in the lake ecosystem over 100 years, with an input of 1 mol inorganic phosphorus at year 40.

Scenario 1



Phosphorus in the dead organic state, reached a new steady state at 58.4 years, which is around 18 years after the inorganic phosphorus input.

The new steady state amount is 1.8327, which is also larger than the original steady state value.

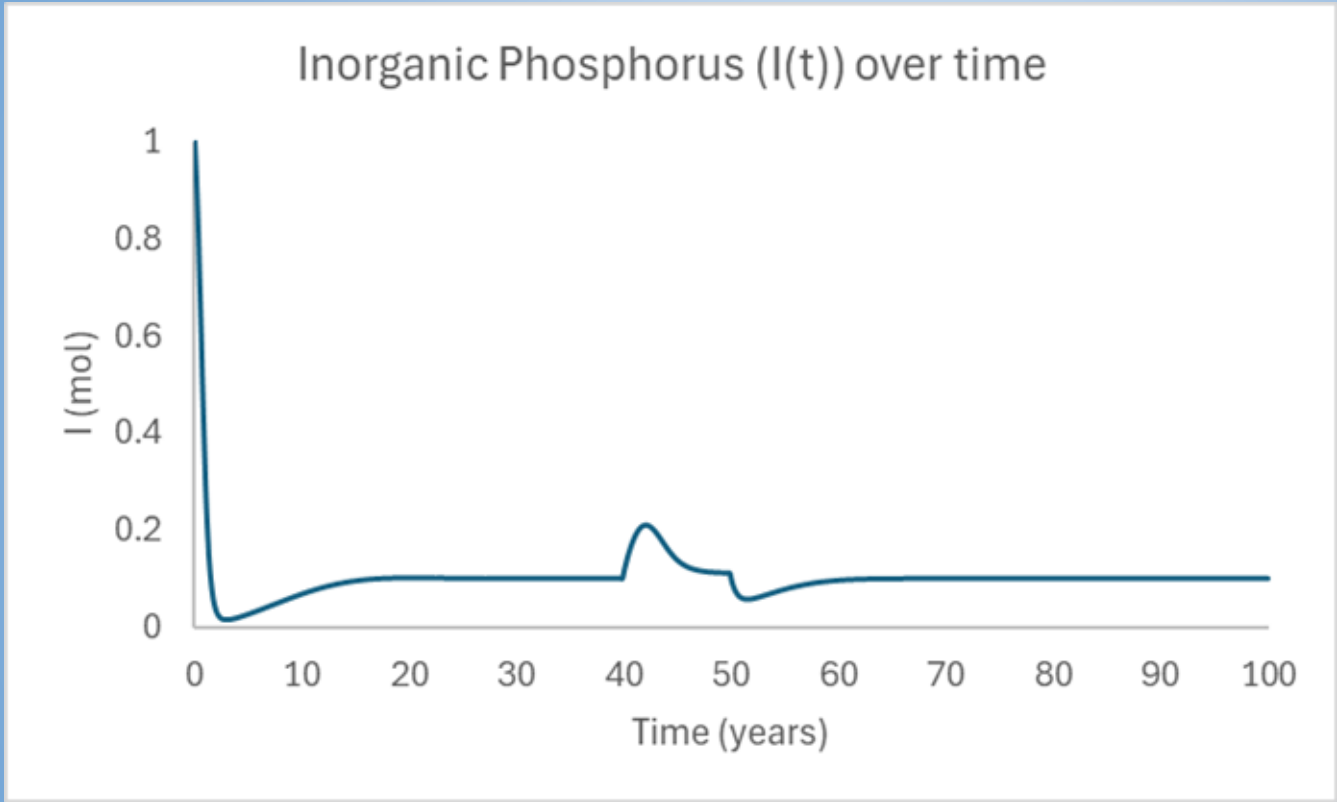
Figure 9. Phosphorus in the dead organic state in the lake ecosystem over 100 years, with an input of 1 mol inorganic phosphorus at year 40.

Scenario 2

0.1 mol of inorganic phosphorus is added to the system at each time interval of 0.1 years starting from year 40 to year 50.

In total: 1 mol of phosphorus added to the system over 10 years.

Scenario 2

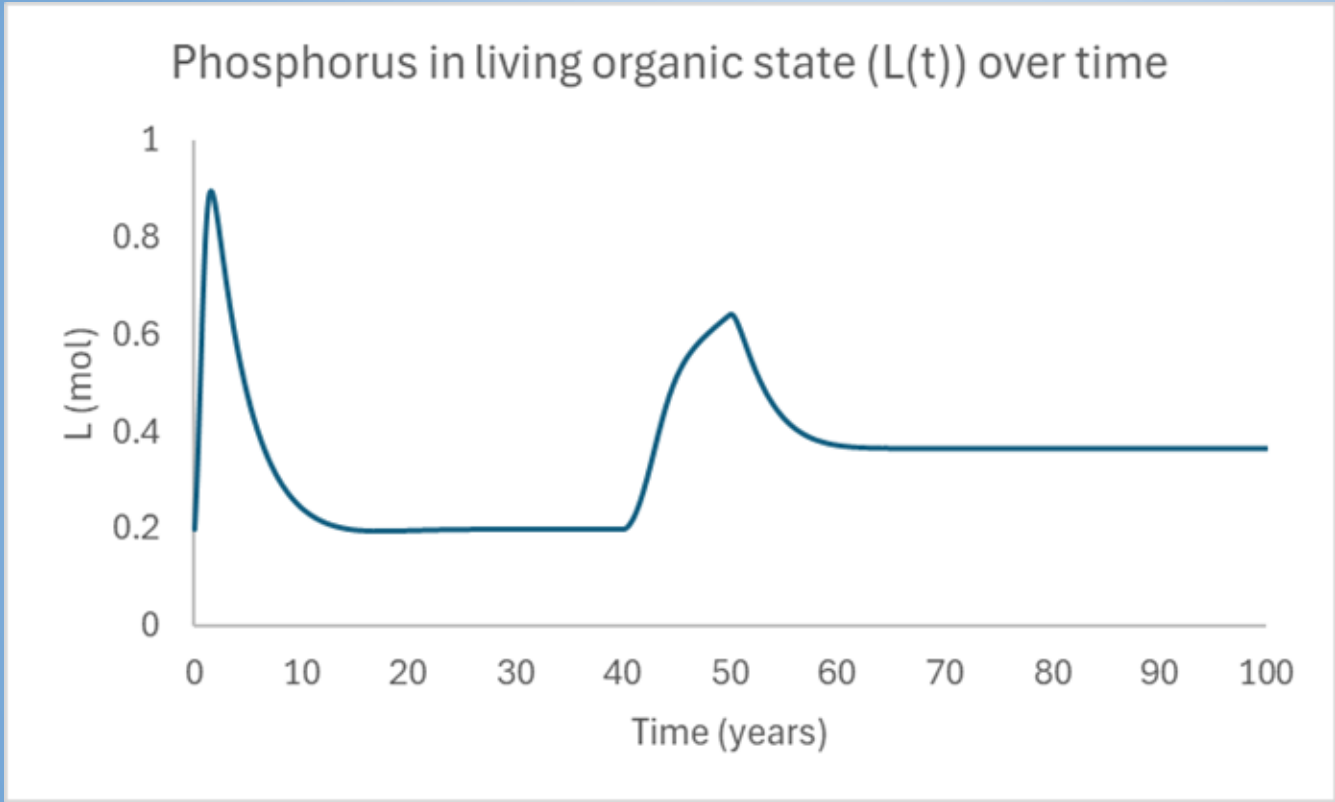


Inorganic phosphorus reaches steady state again in year 66.7, which is around 17 years after the input of phosphorus stops.

The steady state amount of 0.1008 mol is the same as the original steady state.

Figure 10. Inorganic phosphorus in the lake ecosystem over 100 years, with an input of 1 mol of inorganic phosphorus steadily added over the years 40 to 50.

Scenario 2

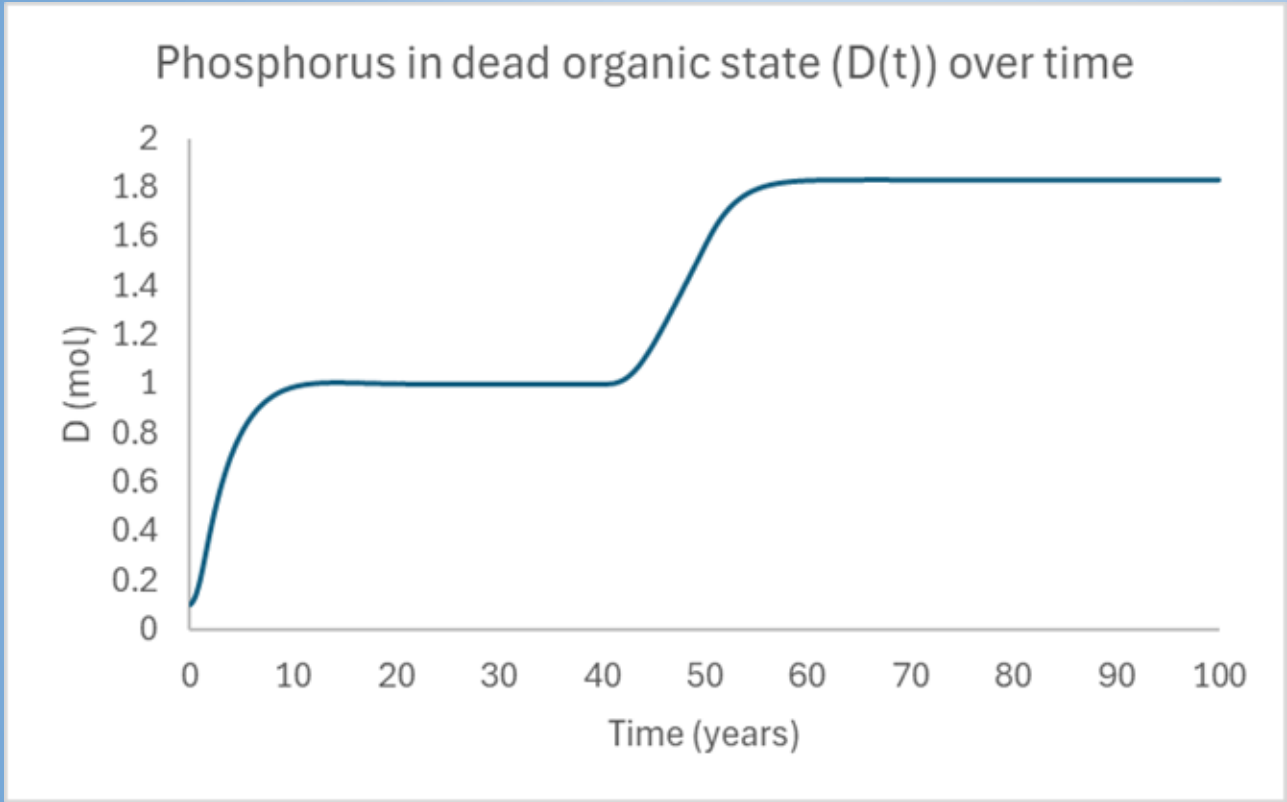


Phosphorus in living organic state, reached a new steady state in year 66.4, meaning it took around 16 years to reach a new steady state after there stopped being an input in inorganic phosphorus.

The new steady state amount is 0.3665 mol, which is greater than the original steady state amount.

Figure 11. Phosphorus in living organic state in the lake ecosystem over 100 years, with an input of 1 mol of inorganic phosphorus steadily added over the years 40 to 50.

Scenario 2



Phosphorus in the dead organic state, it reached a new steady state in year 64.5, meaning it took around 15 years to reach a new steady state after the inorganic phosphorus input stopped.

The new steady state amount is 1.8327 mol, which is greater than the original steady state amount.

Figure 12. Phosphorus in dead organic state in the lake ecosystem over 100 years, with an input of 1 mol of inorganic phosphorus steadily added over the years 40 to 50.

Conclusion: Scenarios

Compare scenario 1 and 2:

- Regardless of whether the inorganic phosphorus is added in one step or over a span of years, the final steady states of the reservoirs would be the same.
- The time it takes for the new steady states to be reached is different.
- For both scenarios, the new steady state amount and the original steady state amount remained the same for the inorganic phosphorus reservoir
- For both the phosphorus in dead organic matter and phosphorus in living organic matter reservoirs, the new steady state amount was greater than the original steady state amount.

Conclusion: Successes, limitations and applications

Success:

- Connects the three phosphorus reservoirs and provides rough estimates of how each of them change over time

Limitation:

- Our model only considers phosphorus in the lake ecosystem
- The actual ecosystem has many other nutrients or factors that likely have an impact and would be affected by the industrial waste

Broader applications:

- Support construction planning
- Convey importance of understanding the implications that modern infrastructure development can have
- Build away from lakes

Appendix

Member Contributions:

Alexis Kuerbig: Slide editing/organization, introduction/background research

Rachel Leung: Slide editing, part 2 models

Niko Vahlas: Slide editing, video editing, part 2 models

Cindy Wong: Slide editing, organization, model description

Ruoxuan Li: Slide editing & organization, part 2 models